

Singular anharmonicities and the analytic continued fractions.

II. The potentials $V(r) = ar^2 + br^{-4} + cr^{-6}$

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The $c = 0$ results of Paper I [J. Math. Phys. 30, 23 (1989)] are extended. In spite of the presence of an additional coupling constant, the Laurent series solutions of the Schrödinger equation that are obtained remain similar to Mathieu functions. Indeed, the recurrences for coefficients preserve their three-term character, their analytic continued fraction solutions still converge, etc. The formulas become even slightly simpler for $c \neq 0$ due to a certain symmetry of the equations to be solved. An acceleration of convergence is better understood and a few numerical illustrations of efficiency are also delivered.

I. INTRODUCTION

For the class of potentials

$$V(r) = ar^2 + br^{-4} + cr^{-6}, \quad a > 0, \quad c > 0, \quad (1.1)$$

the radial Schrödinger bound-state problem

$$\left[-\frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} + V(r) \right] \psi(r) = E\psi(r),$$

$$l = 0, 1, \dots, \quad \psi \in L_2(0, \infty), \quad (1.2)$$

may easily be solved numerically, by some standard (say, Runge-Kutta¹) method. Indeed, in both the threshold and asymptotic domains, the required logarithmic derivatives have a simple WKB form:

$$\psi'(r_0)/\psi(r_0) = 2\lambda r_0^{-3} + O(r_0^{-1}),$$

$$\lambda = \frac{1}{2}\sqrt{c} > 0, \quad r_0 \ll 1, \quad (1.3)$$

and

$$\psi'(r_\infty)/\psi(r_\infty) = -2\mu r_\infty + O(r_\infty^{-1})$$

$$\mu = \frac{1}{2}\sqrt{a} > 0, \quad r_\infty \gg 1, \quad (1.4)$$

respectively. Nevertheless, a few extremely interesting features of the interaction (1.1) emerge after its deeper non-numerical analysis.

(A) The finite polynomial interpolation between $r = 0$ (1.3) and $r = \infty$ (1.4), namely,

$$\psi(r) = r^\kappa \exp(-\mu r^2 - \lambda r^{-2}) \sum_{m=-M}^N h_m r^{2m}, \quad (1.5)$$

may coincide with the exact bound states for certain values of the couplings.²

(B) In the infinite "polynomial-approximation" $N \rightarrow \infty$ and $M \rightarrow \infty$ limit, the Laurent-series Ansatz (1.5) remains useful: It converts our second-order differential equation (1.2) into its second-order difference-equation counterpart:

$$A_{n+1}h_n + B_{n+1}h_{n+1} + C_{n+1}h_{n+2} = 0,$$

$$A_{n+1} = 2\mu(4n + 2\kappa + 1) - E,$$

$$B_{n+1} = -(2n + \kappa + 1)(2n + \kappa + 2) + l(l+1) + 8\mu\lambda,$$

$$C_{n+1} = -2\lambda(4n + 2\kappa + 5) + b,$$

$$n = \dots, -1, 0, 1, \dots \quad (1.6)$$

This motivated the present paper: In the spirit of its preced-

ing part I (Ref. 3) [devoted to the $c = 0$ special case of (1.1) where only property (B) holds], we shall show how our transition to (1.6) simplifies the original problem (1.2). In Sec. II, we relate the "redundant" $n \rightarrow \infty$ boundary conditions to the "Floquet" parameter⁴ κ . In the improved notation of Sec. III, we then define the continued fraction solutions to (1.6) and describe a new type of acceleration of their convergence. Finally, we illustrate and summarize the resulting bound-state prescription in Secs. V and VI, respectively.

II. THE DIFFERENCE SCHRÖDINGER EQUATION

A. The $r \rightarrow 1/r$ symmetries

The form of our differential equation (1.2) remains unchanged after the reflection of coordinates $r \rightarrow \hat{r}$, $\psi \rightarrow \hat{\psi}$, i.e., say,

$$\hat{r} = 1/r, \quad \hat{\psi}(\hat{r}) = \psi(r)/r, \quad (2.1a)$$

which implies merely the change of notation

$$\hat{l} = l, \quad \hat{a} = c, \quad \hat{b} = -E, \quad \hat{c} = a, \quad \hat{E} = -b. \quad (2.1b)$$

The related interchange of asymptotics (1.3) and (1.4) may be characterized by an introduction of the "capped" symbols

$$\hat{\mu} = \lambda, \quad \hat{\lambda} = \mu. \quad (2.1c)$$

Moreover, we may put $\hat{n} = -n$, denote

$$\hat{A}_{\hat{n}} = C_n, \quad \hat{B}_{\hat{n}} = B_n, \quad \hat{C}_{\hat{n}} = A_n,$$

$$\hat{\kappa} = 1 - \kappa, \quad \hat{h}_{\hat{n}} = h_n, \quad (2.1d)$$

and extend the same formal symmetry to our difference Schrödinger equation (1.6). As a consequence, a transition $n \rightarrow -n$ may formally be interpreted as mere "capping" transformation of the parameters.

B. The $|n| \rightarrow \infty$ asymptotics

The leading-order asymptotical form of (1.6) reads

$$8\mu n h_n - 4n^2 h_{n+1} - 8\lambda n h_{n+2} = 0, \quad |n| \gg 1, \quad (2.2)$$

and admits the two independent "Jost" solutions $h_n^{(i)}$, $i = 1, 2$, with

$$h_{n+1}^{(1)}/h_n^{(1)} = 2\mu/n + O(n^{-2}) \quad (2.3)$$

and

$$h_{n+2}^{(2)}/h_{n+1}^{(2)} + -n/2\lambda + O(1), \quad |n| \gg 1. \quad (2.4)$$

Obviously, the "large" $C(n^{|n|})$ components contradict the $M, N \rightarrow \infty$ convergence of our Laurent series (1.5), so that we must demand

$$h_n \approx h_n^{(1)}, \quad n \gg 1, \quad (2.5)$$

and

$$h_{-m} \approx h_{-m}^{(2)}, \quad m \gg 1. \quad (2.6)$$

The latter two boundary conditions do not replace the physical requirements (1.3) and (1.4): In light of the Floquet theory,⁴ they merely fix the values of the "additional" free parameters κ .

III. SCHRÖDINGER EQUATION AS RECURRENCES

A. The improved asymptotics

An insertion of the asymptotics (2.3) and (2.4) in the difference Schrödinger equation (1.6) or (2.2) recovers its asymptotical degeneracy to the two-term recurrences. Thus, for the positive or negative large n , respectively, the third or first item in (1.6) or (2.2) represents just a small correction of the order $O(n^{-2})$.

In the former case, the *Ansatz*

$$h_{n+1} = \frac{(2\mu)^n \Gamma(n + \alpha + 1)}{\Gamma(n + \beta + 1) \Gamma(n + \gamma + 1)} p_n, \quad (3.1)$$

with

$$\alpha = \frac{1}{2}\kappa + \frac{1}{4} - E/8\mu, \quad \beta, \gamma = \frac{1}{2}\kappa + \frac{3}{4} \pm \frac{1}{2}U, \quad (3.2)$$

$$U^2 = (l + \frac{1}{2})^2 + 8\mu\lambda,$$

improves the first boundary condition (2.5) by a few further corrections. It also simplifies the difference Schrödinger equation which, in the new notation, acquires the form

$$p_{n-1} - p_n = \varphi_n p_{n+1},$$

$$\varphi_n = 4\mu\lambda \frac{(n + \alpha + 1)(n + \delta)}{(n + \beta)(n + \gamma)(n + \beta + 1)(n + \gamma + 1)},$$

$$\delta = \frac{1}{2}\kappa + \frac{3}{4} - b/8\lambda. \quad (3.3)$$

Similarly, the second boundary condition (2.6) finds an adequate gamma-function representation in the alternative *Ansatz*

$$h_{-m-1} = \frac{(2\lambda)^m \Gamma(m + s + 1)}{\Gamma(m + v + 1) \Gamma(m + w + 1)} q_m, \quad (3.4)$$

with the parameters

$$s = -\frac{1}{2}\kappa + \frac{3}{4} + b/8\lambda, \quad w, v = -\frac{1}{2}\kappa + \frac{5}{4} + \frac{1}{2}U. \quad (3.5)$$

Obviously, the related new difference Schrödinger equation

$$q_{m-1} - q_m = \psi_m q_{m+1},$$

$$\psi_m = 4\mu\lambda \frac{(m + s + 1)(m + \epsilon)}{(m + v)(m + w)(m + v + 1)(m + w + 1)},$$

$$\epsilon = -\frac{1}{2}\kappa + \frac{5}{4} + E/8\mu, \quad (3.6)$$

remains related to (3.3) by the capping transformations (2.1), with

$$\hat{p}_n = q_n, \quad \hat{q}_m = p_m, \quad \hat{\varphi}_n = \psi_n, \quad \hat{\psi}_m = \varphi_m, \quad (3.7)$$

and

$$\hat{\alpha} = s, \quad \hat{\beta} = v, \quad \hat{\gamma} = w, \quad \hat{\delta} = \epsilon. \quad (3.8)$$

Thus, without any loss of generality, we may work with one of the two limits $|n| \rightarrow \infty$ only.

B. The initial values

Strictly speaking, the boundary conditions (2.5) and (2.6) need not necessarily be postulated in a capping-symmetric manner. Indeed, both these respective requirements are easily seen to represent just the conditions $p_n \approx \text{const}$ and $q_m \approx \text{const}$ for n and $m \gg 1$. In light of Eqs. (3.3) and (3.6) we may replace them immediately by an alternative requirement

$$p_{N+1} = 0, \quad q_{M+1} = 0, \quad N \rightarrow \infty, \quad M \rightarrow \infty. \quad (3.9)$$

Equation (3.9) admits arbitrary normalizations (say, $p_N^{(1)} = 1$ and $q_M^{(2)} = 1$) and we may then use Eqs. (3.3) and (3.6), respectively as recurrences. In such an interpretation (see, also, Paper I), the respective alternatives (3.3) (with $n \gg 0$) and (3.6) (with $m \gg 0$) are equivalent to the $n \gg 0$ and $n \ll -2$ rows of our original Schrödinger difference equation (1.6), and define the "asymptotically correct" (equivalently, "Jost-like") sequences $p_{N-1}^{(1)}, p_{N-2}^{(1)}, \dots, p_{-1}^{(1)}$ and $q_{M-1}^{(2)}, q_{M-2}^{(2)}, \dots, q_{-1}^{(2)}$, respectively. The remaining $n = -1$ row,

$$A_0 h_{-1}^{(2)}/h_0^{(2)} + B_0 + C_0 h_1^{(1)}/h_0^{(1)} = 0, \quad (3.10)$$

may be employed as a matching condition for the two recurrences. Computationally, the latter equation restricts only our freedom in choosing parameters: In what follows, we shall use it as a definition of κ 's for each choice of energy E .

IV. THE ANALYTIC CONTINUED FRACTIONS AND AN ACCELERATION OF THEIR CONVERGENCE

We may introduce the quantities

$$f_n = p_n/p_{n-1}, \quad g_m = q_m/q_{m-1}, \quad (4.1)$$

and rewrite our Schrödinger equation [recurrences (3.3) and (3.6)] in the form

$$f_n = 1/(1 + \varphi_n f_{n+1}), \quad g_m = 1/(1 + \psi_m g_{m+1}) \quad (4.2)$$

In combination with (3.9), they represent the standard analytic continued fractions⁵⁻⁷

$$f_n = \frac{1}{1 + \varphi_n \frac{1}{1 + \varphi_{n+1} \frac{1}{1 + \dots}}},$$

$$g_m = \frac{1}{1 + \psi_m \frac{1}{1 + \psi_{m+1} \frac{1}{1 + \dots}}}. \quad (4.3)$$

With the finite truncation parameters and initial values

$$f_{N+1} = 0, \quad g_{M+1} = 0, \quad (4.4)$$

our recurrences (4.2) still define good approximants since the corresponding coefficients are asymptotically very small,

$$\varphi_n = O(1/n^2), \quad \psi_m = O(1/m^2). \quad (4.5)$$

Numerically, it is therefore reasonable to define our solutions (3.1) and (3.4) as products, with

$$p_n = f_n f_{n-1} \cdots f_{n_0+1} p_{n_0}, \quad q_m = g_m g_{m-1} \cdots g_{m_0+1} q_{m_0}, \quad (4.6)$$

where p_{n_0} and q_{m_0} represent the two alternative normalizations.

In general, the continued fractional convergence may be accelerated by means of the fixed-point technique as described in Paper I. An alternative rearrangement will be described now—it seems to be much more efficient for the particular form of our recurrences (4.2) [with the small parameters (4.5)]. For the sake of brevity, we shall recall the capping symmetry (3.7) and restrict our attention to f_n 's in what follows.

Starting from the slightly modified form of Eq. (3.3),

$$p_{n+1} = \chi_{n+1}^{(0)} p_n - \varphi_{n+1}^{(0)} p_{n+2}, \quad (4.7)$$

$$\chi_{n+1}^{(0)} = 1, \quad \varphi_{n+1}^{(0)} = \varphi_{n+1},$$

we shall eliminate p_{n+1} from Eq. (3.3). We get the $j=1$ formula

$$p_n = \chi_n^{(j)} p_{n-1} - \varphi_n^{(j)} p_{n+j+1}, \quad (4.8)$$

where

$$\chi_n^{(1)} = \frac{\chi_n^{(0)}}{1 + \varphi_n^{(0)} \chi_{n+1}^{(0)}} = \frac{1}{1 + \varphi_n}, \quad (4.9)$$

$$\varphi_n^{(1)} = -\frac{\varphi_n^{(0)} \varphi_{n+1}^{(0)}}{1 + \varphi_n^{(0)} \chi_{n+1}^{(0)}} = -\frac{\varphi_n \varphi_{n+1}}{1 + \varphi_n}.$$

Next, we modify the preceding $j=0$ and $j=1$ formulas,

$$p_{n+2} = \chi_{n+2}^{(0)} p_{n+1} - \varphi_{n+2}^{(0)} p_{n+3}, \quad (4.7')$$

$$p_{n+1} = \chi_{n+1}^{(1)} p_n - \varphi_{n+1}^{(1)} p_{n+3},$$

eliminate p_{n+1} and p_{n+2} from (4.8) with $j=1$, and get the explicit $j=2$ form of the latter prescription, with

$$\chi_n^{(2)} = \frac{\chi_n^{(1)}}{1 + \varphi_n^{(1)} \chi_{n+2}^{(0)} \chi_{n+1}^{(1)}} = \frac{1 + \varphi_{n+1}}{1 + \varphi_n + \varphi_{n+1}},$$

$$\varphi_n^{(2)} = -\varphi_n^{(1)} \frac{\varphi_{n+2}^{(0)} + \chi_{n+2}^{(0)} \varphi_{n+1}^{(1)}}{1 + \varphi_n^{(1)} \chi_{n+2}^{(0)} \chi_{n+1}^{(1)}} = -\frac{\varphi_n \varphi_{n+1} \varphi_{n+2}}{1 + \varphi_n + \varphi_{n+1}}. \quad (4.9')$$

For the subsequent j 's, the coefficients may also be found in the same manner.

In combination with definition (4.1), our relation (4.8) implies that

$$f_n = \chi_n^{(j)} / (1 + \varphi_n^{(j)} f_{n+1} f_{n+2} \cdots f_{n+j+1}). \quad (4.10)$$

These recurrences have the structure of the "extended" continued fractions⁸ and degenerate back to the ordinary continued fractions and Eq. (4.2) for $j=0$. Since $\varphi_n^{(j)} = O(n^{-2j-2})$ is extremely small for large n 's, even the drastic truncations of (4.10), e.g.,

$$f_n = \chi_n^{(j)} (1 + O(1/n^{2j+2}))$$

$$= \chi_n^{(j)} / (1 + \varphi_n^{(j)} \chi_{n+1}^{(j)} \cdots \chi_{n+j+1}^{(j)}) (1 + O(1/n^{4j+4})), \quad (4.11)$$

etc., represent the extremely efficient approximants.

The first nontrivial formulas (4.11) may already be de-

rived from the explicit equation (4.10),

$$f_n = 1 / (1 + \varphi_n - \varphi_n \varphi_{n+1} f_{n+1} f_{n+2}). \quad (4.12)$$

Its simplicity is, surprisingly, preserved also by the higher-order formulas

$$f_n = \frac{Q_{n+1}^{(j-1)}}{\left(Q_n^{(j)} - \prod_{m=n}^{n+j} (-\varphi_m f_{m+1}) \right)}, \quad (4.13)$$

where

$$Q_n^{(j-1)} = Q_n^{(0)} = 1 \quad (4.14)$$

and

$$Q_n^{(j+1)} = Q_{n+1}^{(j)} + \varphi_n Q_{n+2}^{(j)}, \quad j=0,1,\dots, \quad (4.15)$$

in general. The proof by induction is a simple consequence of the identity $(a+b)^{-1} = a^{-1} + a^{-1}b(a+b)^{-1}$. A rearrangement

$$Q_n^{(j)} = 1 + \varphi_n + \sum_{l=0}^{j-2} \varphi_{n+l+1} Q_n^{(l)}, \quad j=2,3,\dots, \quad (4.16)$$

of (4.15) quickly gives the final algorithm for each reasonable j .

V. THE PHYSICAL BOUND STATES

A. An imposition of boundary conditions

In a way paralleling Paper I, we may pick up two roots $\kappa_1(E)$ and $\kappa_2(E)$ of the "Hill determinant" [Eq. (3.10)] or of its equivalent continued fractional (or extended continued fractional) form

$$1 + \psi_{-1} g_0 + \psi_{-1} f_0 = 0. \quad (5.1)$$

The corresponding pair of the Laurent-series solutions $\psi^{(i)}(r) = \psi(r, \kappa_i)$, $i=1,2$, may be then used in a superposition

$$\psi(r) = d_1 \psi^{(1)}(r) + d_2 \psi^{(2)}(r) \quad (5.2)$$

and defines the general solution of our differential equation (1.2) of the second order. We have to avoid the cases where the two functions $\psi^{(i)}(r)$ are linearly dependent (e.g., for $\kappa_1 = \kappa_2 + \text{an even integer}$).

In accord with the standard oscillation theorems,⁴ the number of nodes (zeros) of $\psi(r)$ may only increase with the increasing energy E . As a consequence (see, also, Paper I), all the binding energies may be characterized by an asymptotic emergence of a node in $\psi(r)$,

$$\psi(r_0) = 0, \quad \psi(r_\infty) = 0, \quad r_0 \ll 1, \quad r_\infty \gg 1. \quad (5.3)$$

In the limit $r_0 \rightarrow 0$ and $r_\infty \rightarrow \infty$, such a pair of equations will specify the exact binding energies and coefficients d_i in (5.2) in principle. Thus an insertion of (5.2) gives

$$d_1 \psi^{(1)}(r_0) + d_2 \psi^{(2)}(r_0) = 0,$$

$$d_1 \psi^{(1)}(r_\infty) + d_2 \psi^{(2)}(r_\infty) = 0, \quad (5.4)$$

and the related secular equation

$$\det \begin{pmatrix} \psi^{(1)}(r_0) & \psi^{(2)}(r_0) \\ \psi^{(1)}(r_\infty) & \psi^{(2)}(r_\infty) \end{pmatrix} = 0 \quad (5.5)$$

defines the spectrum of energies.

In place of the simple-minded requirement (5.3), we may also employ our knowledge of the more precise asymp-

otics (1.3) and (1.4). This leads to a pair of equations similar to (5.4),

$$\begin{aligned} d_1 A_{10} + d_2 A_{20} &= 0, \\ d_1 A_{1\infty} + d_2 A_{2\infty} &= 0, \end{aligned} \quad (5.6)$$

where

$$\begin{aligned} A_{ij} &= \sum_{m=-\infty}^{\infty} h_m(\kappa = \kappa_i) r^{2m + \kappa_i} (2m + \kappa_i + \Delta_j), \\ i &= 1, 2, \quad j = 0, \infty, \end{aligned} \quad (5.7)$$

and the constants Δ_j represent the explicit second-order WKB corrections

$$\Delta_0 = -\frac{3}{2} - b/4\lambda, \quad \Delta_\infty = \frac{1}{2} - E/4\mu. \quad (5.8)$$

The energies may also be computed as roots of the modified secular equation

$$\det \begin{pmatrix} A_{10} & A_{20} \\ A_{1\infty} & A_{2\infty} \end{pmatrix} = 0 \quad (5.9)$$

in principle.

B. Numerical examples

For a given set of couplings in our Schrödinger equation [(1.1) and (1.2)], we may employ Eq. (5.1) as an implicit definition of functions $\kappa_1(E)$ and $\kappa_2(E)$, and determine numerically the binding-energy root from Eq. (5.5). The former procedure resembles a search for solutions of quadratic equations: We have chosen our examples in such a manner that the two roots happen to be complex conjugated,

$$\kappa_{1,2} = \text{Re } \kappa \pm i\omega(E).$$

We have also scaled $r \rightarrow \text{const} \times r$ in such a way that $a = 1$. Then, we have employed the purely numerical Runge-Kutta¹ algorithm (cf., also Paper I) and determined the "reference" exact energies E_{exact} . Their sample is listed in Table I for $c = 0.8$ and for a few different values of b . For our choice of the matching points (quite close to the semiclassical turning points: $r_0^{(\text{WKB})} \doteq 0.74$ for $E = 4.8$, and $r_0^{(\text{WKB})} \doteq 0.66$ for $E = 9.44$, etc.), the ground-state precision is already quite satisfactory. For the first excited state, high precision is not yet achieved.

The first comparison of roots E of Eq. (5.5) with E_{exact} is displayed in Table II. Again, very good precision is

TABLE I. The convergence of energies with $r_0 \rightarrow 0$ and $r_\infty \rightarrow \infty$ (the standard Runge-Kutta method, $a = 1$, $c = 0.8$): (A) ground states, (B) the first excited states.

	b		1.00	1.02	1.04	1.06
	r_0	r_∞	E_{exact}			
(A)	0.3	3.5	4.876 70	4.883 68	4.890 63	4.897 54
	0.25	4.0	4.875 98	4.882 95	4.889 89	4.896 80
	0.2	4.5	4.875 93	4.882 94	4.889 86	4.896 77
(B)	0.3	3.5	9.441 4	9.448 7	9.456 0	9.463 1
	0.25	4.0	9.378 7	9.385 7	9.392 6	9.399 5
	0.2	4.5	9.373 6	9.380 6	9.387 5	9.394 4

TABLE II. A comparison of present results [E from Eq. (5.5)] with E_{exact} of Table I. (A) $E - E_{\text{exact}}$, ground states. (B) $E - E_{\text{exact}}$, the first excited states.

	b		1.00	1.02	1.04	1.06
	r_0	r_∞	$E - E_{\text{exact}}$			
(A)	0.3	3.5	0.007 30	0.007 34	0.007 39	0.007 43
	0.25	4.0	0.000 370	0.000 373	0.000 376	0.000 379
	0.2	4.5	0.000 016	0.000 016	0.000 016	0.000 016
(B)	0.3	3.5	0.063	0.520	0.185	0.021
	0.25	4.0	0.023	0.023	0.023	0.023
	0.2	4.5	0.001 3	0.001 3	-0.000 5	-0.064

achieved for ground states, and good results also appear in the excited states. An exception (the last excited-state column) may immediately be understood after an inspection of Table III where the corresponding κ 's are listed: Close to the real axis, the values of $\omega(E)$ remain still strongly sensitive to the variation of r_0 and r_∞ . This represents one of the immanent limitations of the method—in a way analogous to the theory of Mathieu functions,⁹ a modification of the *Ansatz* is needed at the singularity $\omega(E) \rightarrow 0$. Because of the worsening of convergence as observed above, such a situation is less interesting in the present context; vice versa, the increase of $|\omega(E)|$ becomes reflected by an improvement of precision—this may be illustrated by a comparison of Tables II and III in the ground-state cases (A).

A priori, an improvement of the boundary conditions [equal to a transition to the secular equation (5.9)] should also improve the rate of convergence. In practice, it is not always so—this problem appears even in implementations of the Runge-Kutta method sometimes. Here, an explanation is quite easy—the differentiation worsens the rate of convergence of the expansions (5.7). As a consequence of the related rounding errors, Eq. (5.9) is not always superior to Eq. (5.5) in the numerical sense. A typical illustration of this ambivalence is presented in Table IV. Again, the position of singularities is relevant—in accord with Table V, the ground-state root $\kappa_1(E)$ moves towards the real axis during an increase of the coupling b from the value 0.8. In fact, the complex conjugate roots meet [reach $\omega(E) = 0$] somewhere in between the values of $b = 0.904$ and 0.908 . Table V illustrates also the further movement of $\kappa_1(E)$ —presumably, it reaches the value $\kappa_1 = 1$ at $b = 1$.

TABLE III. Parameters $\kappa_{1,2} = 0.5 \pm i\omega$ [= roots of Eq. (5.1)] pertaining to Table II.

	b		1.00	1.02	1.04	1.06
	r_0	r_∞	ω			
(A)	0.30	3.5	0.660 42	0.678 22	0.695 15	0.711 30
	0.25	4.0	0.660 58	0.678 35	0.695 24	0.711 35
	0.20	4.5	0.660 59	0.678 35	0.695 25	0.711 35
(B)	0.30	3.5	0.568	0.651	0.490	0.305
	0.25	4.0	0.534	0.469	0.383	0.253
	0.20	4.5	0.525	0.458	0.367	0.137

TABLE IV. A comparison of results for the simple and WKB boundary conditions [Eq. (5.1) for $\kappa_{1,2} = 1.5 \pm i\omega(E)$ and the respective Eqs. (5.5) and (5.9) for energies]. (A) The ground state with $a = c = 1$ and $b = 0.8$. (B) The first excited state with $a = c = 1$ and $b = 1$.

	Boundary conditions		Simple		WKB	
	r_0	r_∞	$\omega(E)$	$E - E_{\text{exact}}$	$\omega(E)$	$E - E_{\text{exact}}$
(A)	0.3	3.5	0.444 35	0.007 75	0.459	0.044
	0.25	4.0	0.441 05	0.000 44	0.452	0.029
	0.2	4.5	0.440 88	0.000 14	0.447	0.016
(B)	0.3	3.5	1.170 2	0.200	1.142 49	0.0641
	0.25	4.0	1.144 5	0.258	1.141 25	0.0049
	0.2	4.5	1.141 3	0.002	1.141 17	0.0004

VI. CONCLUDING REMARKS

In a way that resembles the use of the exact Mathieu solutions for potentials $V \sim 1/r^4$ (see, e.g., Ref. 10 or the review by Newton¹¹), we have described here the quasi-exact construction of bound states in the potential (1.1).

A core of our construction lies in the use of continued fractions. In this respect, our method and, in particular, the related new technique of acceleration of their convergence, might prove relevant in a broader methodical context. In conclusion, let us mention at least the following three possibilities.

(a) In a number of papers, we may find an analytic continued fraction rearrangement of the three-term recurrences that resemble our Eq. (1.6) even by their power-series origin. In the underlying spectrum of technical problems, we may distinguish between the mathematically “safe” variational tractability of the potentials derived from

$$r^2 + fr^2/(1 + gr^2) \quad (6.1)$$

(see e.g., Ref. 12), and the straightforward (so-called “Hill-determinant”) treatments of the forces of the type

$$ar^2 + br^4 + cr^6 \quad (6.2)$$

(see, e.g., Ref. 13). Both these techniques lie very close to our present construction: The acceleration of the continued-

TABLE V. The first ground-state root $\kappa_1(E)$ of Eq. (5.1). An example of its transition in the complex plane ($a = c = 1$).

r_0	0.30		0.25	
r_∞	3.50		4.00	
b	Re κ_1	Im κ_1	Re κ_1	Im κ_1
1.00	1.0005	0.0	1.0000	0.0
0.98	1.0705	0.0	1.0699	0.0
0.94	1.2259	0.0	1.2251	0.0
0.92	1.3331	0.0	1.3319	0.0
0.908	1.4639	0.0	1.4590	0.0
0.904	1.5000	0.1035	1.5000	0.0843
0.90	1.5000	0.1383	1.5000	0.1247
0.88	1.5000	0.2438	1.5000	0.2367
0.80	1.5000	0.4443	1.5000	0.4410
0.60	1.5000	0.6686	1.5000	0.6669
0.20	-0.5000	-0.8731	-0.5000	-0.8721

fractional convergence plays an important role in it.

(b) Formally, a very broad class of applications of continued fractions has been inspired by the Lanczos ideas¹⁴ related to the theory of moments.⁵⁻⁷ For a sketchy illustration, we may recall, e.g., the diagonalization of the so-called chain models in the condensed-matter physics¹⁵ or the extensive Lanczos-type computations in nuclear physics.¹⁶ Also, we may mention here the closely related rearrangements of the divergent Born series¹⁷ or the whole rich domain of the nonequilibrium statistics,¹⁸ etc. Often, an acceleration of convergence remains an open problem in this context.¹⁹

(c) Whenever we interpret continued fractions as a special case of the Padé resummation,²⁰ we immediately get in close contact with the perturbation theory.²¹ In this setting, an acceleration of convergence is of extreme importance²² and, presumably, new inspiration could stem from the present results. Unfortunately, this question already lies too far beyond the scope of the present paper.

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